Worksheet 8-9 & 8-10 A Volumes: Disc Method #1
AP Calculus Name_____

Directions: Find the volume of the solid obtained by rotating the region bounded by the given curves about the specified line. Sketch the region, the solid, and a typical disc.

1.) $y = x + 1,\ y = 0,\ x = 0,\ x = 2$ about the x-axis

2.) $y = \frac{1}{x}\ ,\ y = 0,\ x = 1,\ x = 4$ about the x-axis

3.) $y = \sqrt{x-1},\ y = 0,\ x = 5$ about the x-axis

4.) $y = e^x,\ y = 0,\ x = -1,\ x = 1$ about the x-axis

5.) $x = 2\sqrt{y},\ x = 0,\ y = 9$ about the y-axis

6.) $2x = y^2,\ x = 0,\ y = 4$ about the y-axis

7.) $y = x^3,\ x = 2,\ x - \text{axis}$ about the line $x = 2$

8.) $y = x^2 + 2,\ y = 3,\ y - \text{axis}$ about the line $y = 3$

Answers:

1.) $\frac{26\pi}{3}$ 2.) $\frac{3\pi}{4}$ 3.) 8π 4.) $\frac{\pi}{2}\left(e^2 - \frac{1}{e^2}\right)$ 5.) 162π 6.) $\frac{256\pi}{5}$

7.) $\frac{16\pi}{5}$ 8.) $\frac{8\pi}{15}$